

Objecthood as a composition of differentiae: scaling the world model with a network of modulating zones in an embodied Sensation Modulating Network

G. Nagarjuna*

Durgaprasad Karnam[†]

June 13, 2026

Abstract

An agent’s grip on its world is usually derived either by training a world-model on data or by inverting an internal generative model. We pursue a third route, from the Sensation Modulating Network (SMN): in a body built as a *network of opponent, pull-only, dual-interface zones* immersed in a real gravitational field, the rudiments of objecthood, agency, and directedness are *enacted by construction*. We make one thesis precise and falsifiable on a physics-engine bench. A single zone registers a *difference* along its line of modulation — a *differentia* — not yet an object: it reports how a thing answers the agent’s own pull (objecthood-as-resistance), factors **self / field / object** through a force-reafference model that predicts its own effort against gravity, and sustains a halt that is persistent, returnable, and side-specific (directedness). An *object* is a *composition of differentiae*, and composition requires a *network* of zones coupled through the modulatory board. We then show the clinching result: the agent’s world-model capacity — the objects it can individuate — scales as 2^K with the number of zones the network binds (verified against real multi-zone presses), while zones present but *unbound* add nothing. The world model is built by the *network*, not by the sensor count: scaling the modulation is scaling the world, and its onset is a *percolation-like transition* in the network’s coupling density. Everything is reproducible from an open bench.

1 Introduction

How does an agent come to have a world of objects it can act on, to tell its own doings from the world’s, and to be *about* something? Two answers dominate. The machine-learning answer trains a world-model from a corpus [4]. The active-inference answer equips the agent with an internal generative model and minimizes prediction error by inferring hidden causes [12]. Both place the structuring prior *inside* the agent.

We pursue a third route, from the SMN architecture [10]. An SMN agent is a network of *opponent, dual-interface zones*: each zone acts through a pull-only antagonist pair and senses, through the same interface, the effort and contact that pull meets. The agent is immersed in a real physical field and reads that field *only through the work it does against it*. On this view the structuring prior is not in the head but in the world–body coupling — gravity is a strong, free prior

*IISER Pune, India.

[†]IIT Bombay, India.

— and the agent’s grip on its world is *enacted by construction*, neither learned from data nor inferred over a model.

Our central claim is a claim about *composition and scale*. A single zone registers a *difference* along one line of modulation — a *differentia* in the classical sense, a “difference that makes a difference” [1]. A difference is not an object. An *object* is a *composition* of differentiae (Aristotle’s genus-plus-differentiae; a convex region across several domains [5]), and to compose differentiae the body must *couple* its zones — it must be a network. This dissolves an obvious objection (“where is the network?”): the body *is* the network, and the world it can construct grows with it. Four results, none requiring any training, develop the claim on a contact bench:

1. **A single zone is a difference register.** Pressing on a thing, one zone reads how it answers the pull — a force→result mapping that separates objects by resistance, the one differentia the zone affords.
2. **Self, field, and object are factored architecturally**, by a force-reaffERENCE model that predicts the agent’s own effort against the field.
3. **Haltability generates aboutness:** that single-zone register can hold a directed, returnable, side-specific halt — the body’s way of being about a thing.
4. **Objecthood is a composition of differentiae, and it scales with the network.** Coupling K zones through the board composes K differentiae into an object; the world-model capacity scales as 2^K with the *bound* network, not with the raw sensor count.

2 Background: differences, forces, and the construction of objects

That objects are *composed* of differences is an old thought made precise in several traditions. Bateson [1] took the elementary unit of mind to be a “difference that makes a difference”; classically an object is individuated by its *differentiae*; and Gärdenfors [5] models an object category as a convex region across *several* quality domains — a composition of differences, one per domain. That objects are not *given* but *constructed* — wrestled from a flux by an engaged subject through what Smith [15] calls *registration*, a deliberate middle course between realism and constructionism — is the metaphysical companion of the same thesis. Our contribution is a minimal, sub-semantic, embodied mechanism for it: differences are registered by single zones, objects by their composition across a coupled network.

The single-zone differentia we report is force-based. Following Gärdenfors’s two-vector model of events [6, 7], an event is an action *force vector* mapped to a *result vector*; friction and gravity enter as *counterforces* [16, 17]. Our zone enacts that mapping — the antagonist pull is the force, the thing’s response is the result, the field is the counterforce — by construction rather than by learning. Recurrent dynamical systems can be *trained* to categorize [2, 9]; we do not train. Active inference [12, 13] locates the prior in a represented model; we locate it in the world–body coupling. Brooks [3] rejected dataset and representation alike but reached a ceiling at higher cognition; we keep the no-dataset commitment while adding the modulation that *binds* differentiae into objects. Morphological computation [14] and the sensorimotor/ecological tradition [8, 11] are allies throughout.

thing	yielding (final angle, rad)	resistance (impulse, N s)	dominant register
free	1.20	0.0	free · working · moving
movable-light	1.06	3.1	object · straining · moving
movable-heavy	0.71	9.3	object · straining · halted
compliant	0.37	13.2	object · straining · halted
fixed	0.13	18.9	object · straining · halted

Table 1: E1: one zone, one differentia — five things as five points on the resistance/yielding arc.

3 The bench: zones, the basal floor, and the board

The experiments run on **smn-lab**, an open MuJoCo bench in which the SMN control architecture drives a physical body [10]. The atom is a **zone**: a degree of freedom driven by a *pull-only antagonist pair* — there are no position (“stepper”) commands; the agent modulates two opposing pulls, whose co-activation is the atomic equilibrium we call a *balance*. Each zone’s **dual interface** reads joint state, the actuator effort (the work the pull performs), and contact. Beneath the zones runs the **basal action pattern generator (BAPG)**: not a *central* pattern generator but the architecture’s distributed, evolutionarily ancient innermost *floor*, an endogenous drive the higher layers gate. The BAPG supplies the continuous modulation that fills the spaces *between* the discrete differentiae, which is why the constructed world is not pixelated: it is generated by coupled dynamical systems, not by linear algebra.

Zones are wired by the **board** (the “balance beam”), the coupling network that lets one zone’s reading enter another’s — the substrate of composition. A *single* zone has a trivial board; objecthood, as a composition of differentiae, requires a non-trivial one. The body’s structure fixes, *a priori*, a finite **register lattice** (Contact $\times$ Effort $\times$ Motion per zone) — the space of distinctions the agent can occupy — and an experiment records which subset it occupies. Gravity is on, the floor frictional, and objects carry mass, fixation, and compliance; the agent reads the field through effort. A single-joint probe confirmed the read separates free, movable, compliant, and fixed targets cleanly (not shown).

4 Results

4.1 E1 — a single zone is a difference register

A zone presses five things differing only in how they answer the pull: *free*, *movable-light*, *movable-heavy*, *compliant*, *fixed*. Each is summarized by the contact *impulse* it returns (resistance) and the angle the limb reaches (yielding). The five fall on a single monotone arc (Figure 1, Table 1) — a force→result mapping that registers one *differentia*, resistance-to-modulation. This is honestly a *difference register*, not yet objecthood: along one line of modulation the zone tells things apart, but it composes nothing. Driven instead by a position (“stepper”) controller commanded to an angle, the body *overpresses what it cannot reach*: on the fixed thing its contact impulse is 57 N s versus the SMN pull’s 19 — the unbounded failure of commanding an angle rather than modulating a force.

4.2 E2 — self, field, and object

A sagittal, gravity-loaded zone lifts against the field. By self-motion the agent calibrates a **force-reaffERENCE** model — the effort its own pull needs to move the limb against gravity alone — using the same reaffERENCE predictor that gives the self/world register elsewhere in the bench, now in the

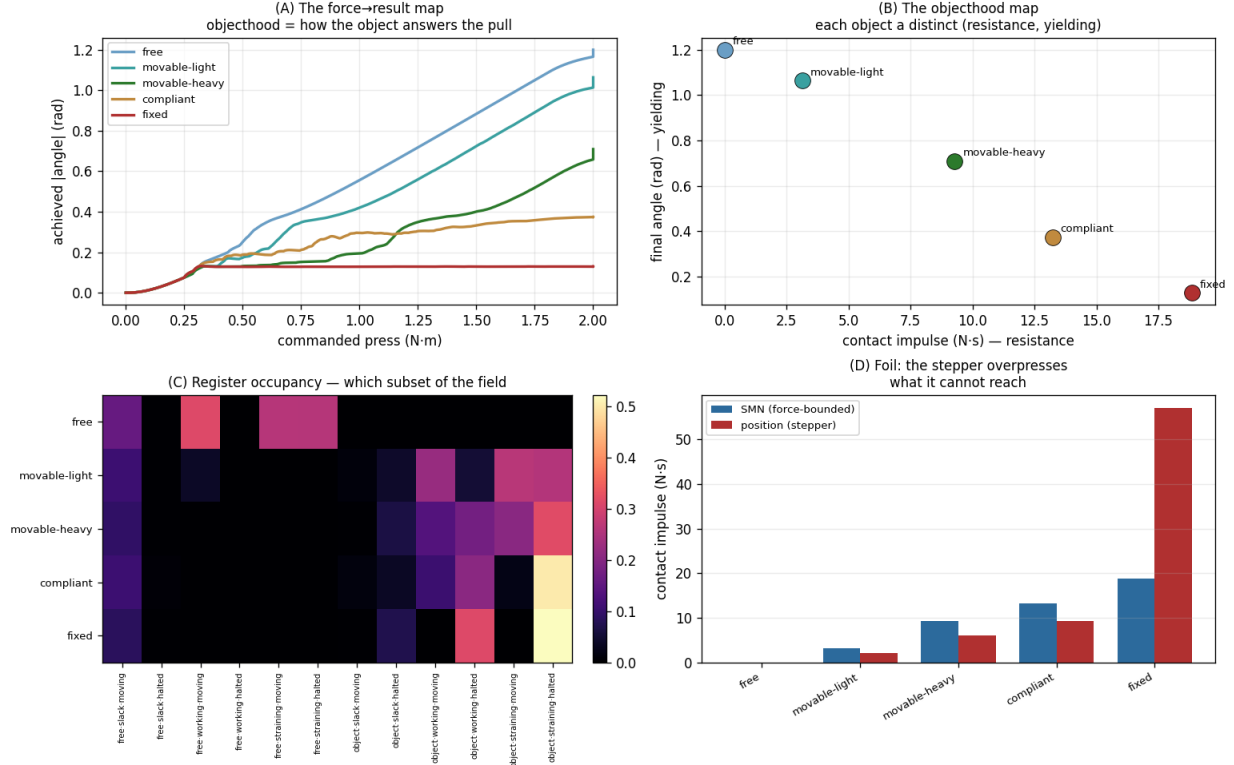


Figure 1: **The single-zone difference register.** The force→result map (A) and the (resistance, yielding) map (B) place five things on one arc; register occupancy is distinct per thing (C); the stepper foil overpresses the unreachable thing while the SMN pull stays force-bounded (D).

force domain. Thereafter the field is a predictable baseline; the residual (effort beyond it) plus contact surface the object, and motion produced with no effort surfaces an external cause. The three conditions are attributed correctly by construction (Figure 2): *self+field* residual ≈ 0.001 , attributed self+field with probability 1.00; *object* attributed 0.68 (its pre-contact phase honestly reading self+field) on a residual of 0.34 plus contact; *external* attributed 1.00. The self/world distinction is thereby generalized into **self / field / object**, with the field — gravity — as the always-present third term that, factored out, never masquerades as an object.

4.3 E3 — haltability generates aboutness

A thing on the left; both limbs receive the same command. A halt-in-contact proves to be a *directed* state, not a stop (Figure 3). It is **persistent** — the left zone sustains a directed halt for 1.51 s and restores it after a perturbation knocks it off; **returnable** — across withdraw/re-press cycles the agent re-acquires the same thing five times; and **side-specific** — the right zone, given the identical command but nothing to press, never holds (0.00 s), so the left-halt/right-free asymmetry *is* directedness. A BAPG-only foil (basal rhythm without the haltability layer) only flickers: its longest sustained halt is 0.26 s (seventeen brief touches), because a rhythm is forced to release. Rhythm is not aboutness; haltability is.

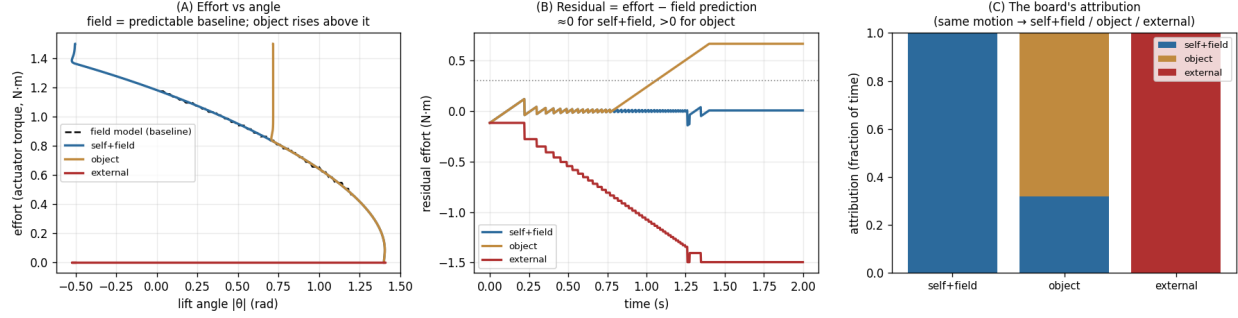


Figure 2: **Self / field / object**. Effort versus angle against the calibrated field baseline (A); the residual beyond the field (B), ≈ 0 for self+field and elevated for the object; the board’s attribution (C).

4.4 E4 — objecthood as composition; the world model scales with the network

A single zone gives a difference; an object is a *composition* of differentiae, which needs a *network*. We make a row of K pull-only antagonist press-zones, each reading one facet’s resistance differentia (hard vs. soft, exactly as in E1), and let the board compose the bound zones into an object code. We then measure the agent’s world-model capacity — the number of things it can individuate — as a function of how many zones the network binds.

The capacity scales as 2^K with the bound network (Figure 4): 2, 4, 8, 16, 32 individuable objects for $K = 1, \dots, 5$, and *every one* of the 62 composed objects is read correctly from real multi-zone presses (per-object accuracy 1.00; per-zone hard impulse 3.83 vs. soft 2.08). Zones that are present but *unbound* add nothing: a single difference, capacity 2, flat. The world model is built by the *network*, not the sensor count — a clean scaling law, $\log_2(\text{capacity}) = (\text{bound zones})$. The “object” is exactly the composition of differentiae the coupled network can hold; the BAPG floor fills the continuum between them, so the constructed world is a smooth dynamical manifold, not a pixel grid.

4.5 E4b — the onset of objecthood is a coupling-density transition

Given reliable per-zone differentiae (E4), *when* do they bind into an object? Model the board as a random graph $G(K, p)$ — each pair of zones coupled with probability p — and let composition bind zones within a connected component, so the richest single object the agent can hold is the largest bound component, of capacity $2^{(\text{largest component})}$. Sweeping the coupling density (Figure 5, $K = 24$ zones), objecthood appears as a **percolation-like transition**: below a critical density the differentiae stay fragmented (the largest bound object a few zones, capacity $\sim 2^4$), and past $p_c = 1/K$ a giant bound object emerges, the capacity jumping to $\sim 2^{18}$ by $p = 2/K$. The order parameter — the largest bound object as a fraction of K — is the classic giant-component curve. Objecthood, in this architecture, is not all-or-nothing: it has a *threshold* in the modulation network’s connectivity. (The per-zone differentiae here are the real bench reads of E4; what is swept is the board’s connectivity. This is the random-board baseline; structured boards and the spectrum of the coupling matrix are the open questions below.)

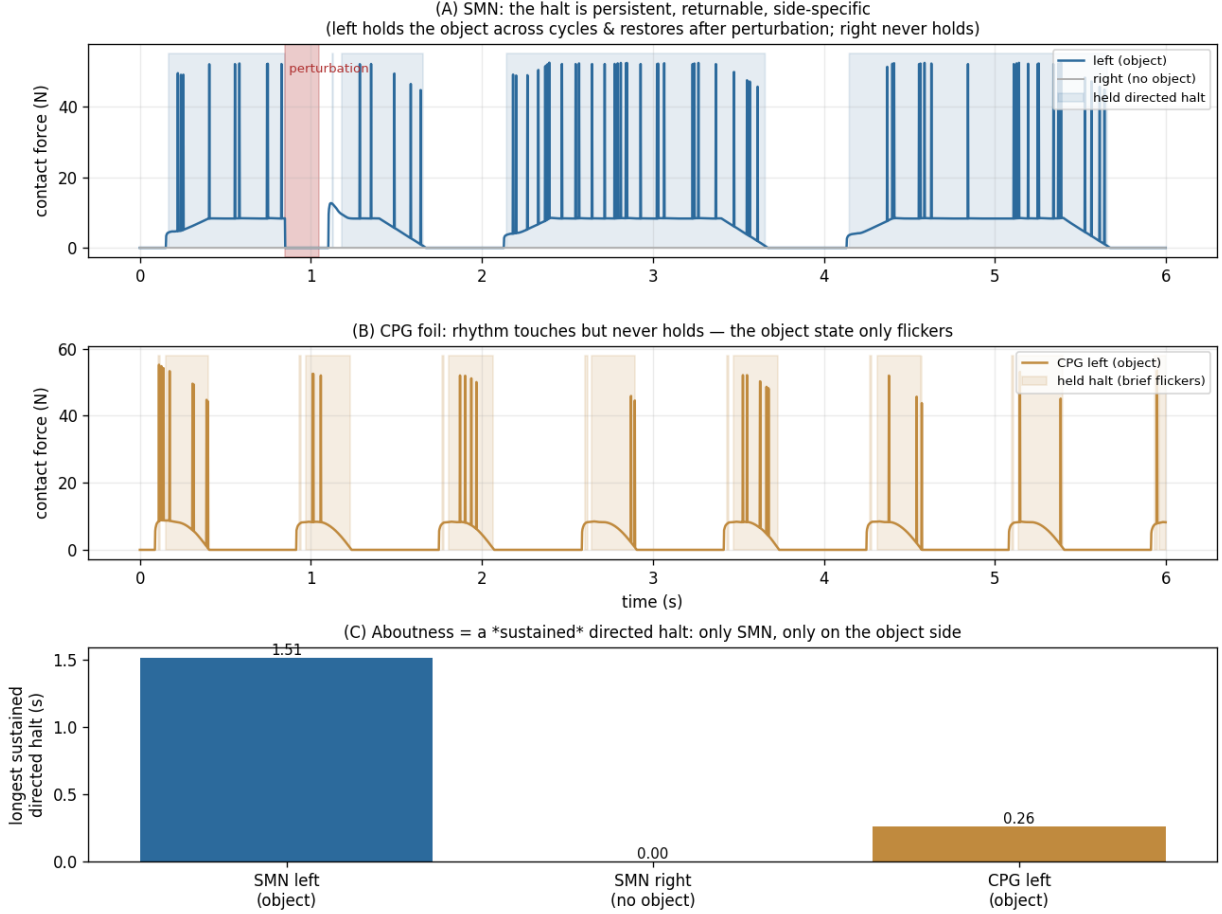


Figure 3: **Haltability generates aboutness.** The halt is persistent, returnable, and side-specific (A); a basal rhythm without the haltable layer can only flicker (B); a sustained directed halt is present only with haltability, only on the object side (C).

5 Discussion

The four results compose into one: *the opponent-modulated body in a field enacts, by construction, what learning or inference would otherwise supply — and the world it constructs scales with the network that binds its differentiae*. A single zone registers a difference (E1) that already carries a self/field/object factoring (E2) and a haltable aboutness (E3); an object is the composition of such differences across coupled zones, and the agent’s world-model capacity grows 2^K with the bound network (E4). Nothing is learned, and nothing inverts an internal model; the structuring prior is the physics the body is coupled to, and the structuring *degrees of freedom* are the couplings of the board.

This places us precisely among neighbours. With Gärdenfors [5, 6] we share the geometry and the two-vector, force-based account of events; we differ in enacting the force→result mapping rather than learning it, and we would offer one refinement of his cognitivism — the *forces* are physically real, while the *objecthood* read off them is constructed. This is Smith’s middle course between

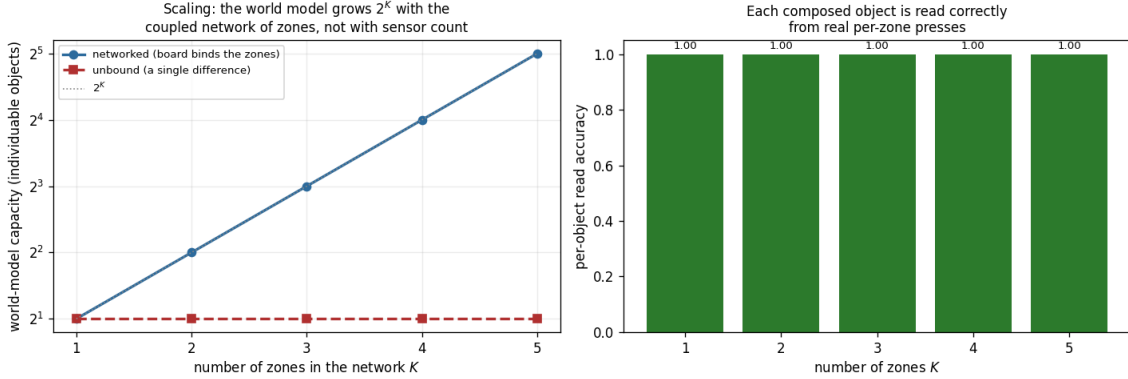


Figure 4: **The world model scales with the network of zones.** World-model capacity grows as 2^K with the coupled network (left), while unbound zones give only a single difference (capacity 2, flat); each composed object is read correctly from real per-zone presses (right). Scaling the modulation is scaling the world.

realism and constructionism in concrete form: the flux is real; the object is wrestled from it by the agent’s modulation, and is exactly as rich as the network can compose. With active inference [12] we share a single principle doing first-principle work; we differ in locating the prior in the world–body coupling — gravity as a free prior, read through the work the agent does against it — rather than in a represented model. Whether “minimize prediction error” and “compensate against the field” are two descriptions of one event is, to us, an open and inviting question.

The scaling result reframes the agent as a **complex network** and opens genuinely physical questions that our cognitive vocabulary cannot answer. E4b already shows the baseline: for a *random* board, objecthood appears at the giant-component threshold $p_c = 1/K$. What remains open is squarely physics: how do *structured* boards — modular, hierarchical, scale-free — shift that onset; *what does the spectrum* of the coupling matrix say about which networks compose objects (a random-matrix question); and what are the dynamics of binding? The bench supplies the coupling matrices and trajectories to ask them. The geometry of the constructed world is itself body-indexed: its dimensionality is set by the network (the number of bound zones), its coordinates by the agent’s own configuration — so morphology fixes the geometry of the world an agent can have.

The account is bounded. The 2^K law is, once each facet is one clean bit, true by construction; the non-trivial content is the *dissociation* (capacity tracks the bound network, not the raw sensor count) and a real multi-zone body reading 62 objects without error. The held halt is supplied by the object plus continued press; a *free-space* co-contraction equilibrium with tunable stiffness needs muscle/tendon actuators (ideal torque motors make co-contraction inert). Unifying the horizontal and gravity-loaded variants on one body, chaining zones into a crawler that builds a habitat-scale model, and the coupling-density transition above, are the next steps. *Naming* — a shared symbol for the composed object — is conventional and social and lies beyond a single agent.

6 Reproducibility

The bench, the units, and an interactive interface are open (github.com/gnowgi/smn-lab, GPL-3.0). Each result is produced by a single seeded script (the E1–E4 experiments) and the figures

E4b --- the world model's objecthood is a percolation-like transition in the board's coupling density ($K=24$ zones)

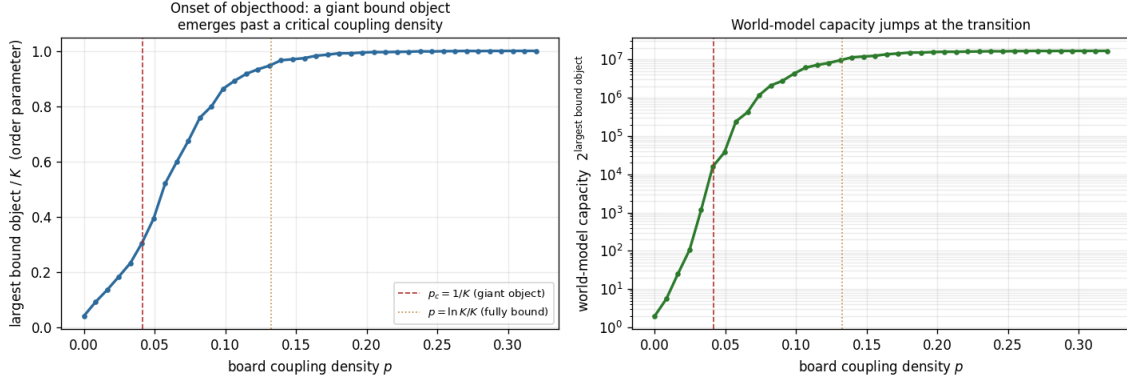


Figure 5: **Objecthood as a percolation transition.** The largest bound object (order parameter, left) and the world-model capacity (right) jump as the board’s coupling density crosses $p_c = 1/K$: below it the differentiae are fragmented, above it a giant bound object emerges.

regenerate from them; the register lattice and the coupling matrix are computable from the body’s structure alone. A versioned snapshot with a Zenodo DOI will accompany the published version.

7 Conclusion

In an embodied Sensation Modulating Network, objecthood is a composition of differentiae: a single zone registers a difference — resistance, with its self/field/object factoring and its halttable aboutness — and an object is what a coupled *network* of zones composes. The world an agent can construct scales with that network, 2^K in the zones it binds, and emerges — as the coupling density rises — at a percolation threshold. Scaling the modulation is scaling the world; the architecture, not a corpus or an inferred model, is doing the work.

References

- [1] Gregory Bateson. *Steps to an Ecology of Mind*. Chandler Publishing, 1972.
- [2] Randall D. Beer. The dynamics of active categorical perception in an evolved model agent. *Adaptive Behavior*, 11(4):209–243, 2003.
- [3] Rodney A. Brooks. Intelligence without representation. *Artificial Intelligence*, 47(1–3):139–159, 1991.
- [4] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [5] Peter Gärdenfors. *The Geometry of Meaning: Semantics Based on Conceptual Spaces*. MIT Press, 2014.
- [6] Peter Gärdenfors. Events and causal mappings modeled in conceptual spaces. *Frontiers in Psychology*, 11:630, 2020. doi: 10.3389/fpsyg.2020.00630.

- [7] Peter Gärdenfors and Massimo Warglien. Using conceptual spaces to model actions and events. *Journal of Semantics*, 29(4):487–519, 2012.
- [8] James J. Gibson. *The Ecological Approach to Visual Perception*. Houghton Mifflin, 1979.
- [9] Wolfgang Maass, Thomas Natschläger, and Henry Markram. Real-time computing without stable states: A new framework for neural computation based on perturbations. *Neural Computation*, 14(11):2531–2560, 2002.
- [10] G. Nagarjuna and Durgaprasad Karnam. The sensation modulating network. *arXiv preprint arXiv:2605.26856*, 2026. URL <https://arxiv.org/abs/2605.26856>.
- [11] J. Kevin O’Regan and Alva Noë. A sensorimotor account of vision and visual consciousness. *Behavioral and Brain Sciences*, 24(5):939–973, 2001.
- [12] Thomas Parr, Giovanni Pezzulo, and Karl J. Friston. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press, 2022.
- [13] Giovanni Pezzulo and Paul Cisek. Navigating the affordance landscape: Feedback control as a process model of behavior and cognition. *Trends in Cognitive Sciences*, 20(6):414–424, 2016.
- [14] Rolf Pfeifer and Josh Bongard. *How the Body Shapes the Way We Think: A New View of Intelligence*. MIT Press, 2006.
- [15] Brian Cantwell Smith. *On the Origin of Objects*. MIT Press, Cambridge, MA, 1996.
- [16] Leonard Talmy. Force dynamics in language and cognition. *Cognitive Science*, 12(1):49–100, 1988.
- [17] Phillip Wolff. Representing causation. *Journal of Experimental Psychology: General*, 136(1): 82–111, 2007.